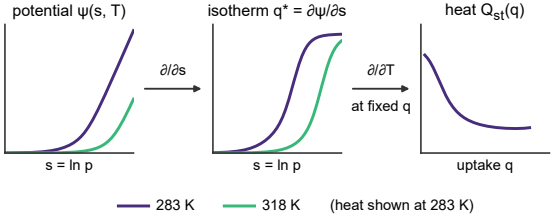
**a** One potential gives the isotherm and the heat (curves illustrative)



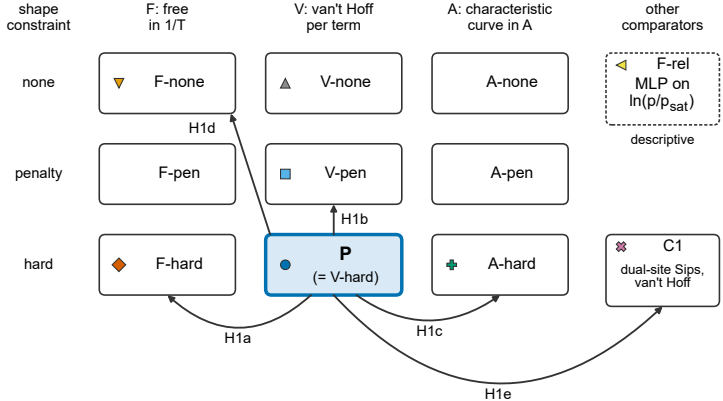

**Holds by construction, for every parameter value**

- monotone in p: $\partial q^*/\partial s = \partial^2\psi/\partial s^2 > 0$, because ψ is convex in s
- Henry limit: q* grows in proportion to p as $p \to 0$
- heat always finite: $Q_{st} = -RT^2\,(\partial q^*/\partial T) / (\partial q^*/\partial s)$, denominator never zero; sign and size not constrained

In P each term has a van't Hoff affinity $K(T) = K_0 \exp(Q / RT)$, Q free in sign.

Every model's heat is computed from its own isotherm by the same code.

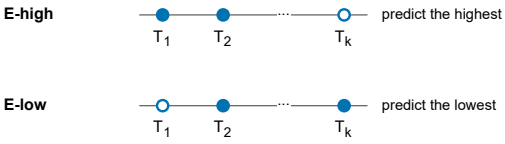
**b** Factorial: shape constraint × temperature structure




**Comparisons of P (H1b primary; H1c-H1e confirmatory, Holm over H1b-H1e; H1a descriptive):**

**H1a** against F-hard: temperature structure
**H1b** against V-pen: hard against penalty
**H1c** against A-hard: characteristic curve
**H1d** against F-none: generic network
**H1e** against C1: classical dual-site Sips

**c** Tasks and scores (per material, temperatures $T_1 < \ldots < T_k$)




scored

**Accuracy (H1, primary)**
E = RMSE / $q_{ref}$ on the held-out isotherm; per material, the mean of $\log_{10}$ E over E-high and E-low

**Violations (H3)**
grid to ±15 K, $p \le 0.9\ p_{sat}$: uptake falling by more than 0.02 $q_{ref}$ as p rises, or below −0.02 $q_{ref}$ (band to ±20 K reported apart)

**Heat (H2, synthetic)**
$Q_{st}$ against the known heat of five synthetic truths, at matched loading in 0.1 to 0.9 $q_{ref}$, on the same tasks